

Estimating latent factors in dynamic models: the Kalman filter

Francesco Zucca, Michael Fehl

Table of contents

Abstract	1
1. The Stock & Watson model	2
1.1. Latent factor estimation	3
1.2. Mapping the latent factor to GDP growth	4
1.3. The normality assumption	4
1.4 Verifying that there is only one underlying factor (TO FINISH)	5
2. Breaking Gaussianity	6
2.1 Gaussian Distribution	6
2.2 Student-t Distribution	7
2.3 Skewed-Elliptical	8
2.4 Monte Carlo Simulation	10
2.5 Efficiency Loss under Non-Gaussian Data	10
3. Two-Factor Model	11
3.1 Introducing More Variables	11
3.2 Estimated Using Stock & Watson	11
3.3 Parallel Analysis	11
Bibliography	11

Abstract

In a 1988 working paper, Stock and Watson proposed a framework to estimate and forecast short-run U.S. economic activity. Their model built on the premise that comovements across multiple macroeconomic variables can be captured by a single, unobserved latent variable. Specifically, they hypothesized that their four chosen indicators form a dynamic one-factor model, where the latent factor represents the state of the business cycle and can therefore be mapped to Gross Domestic Product (GDP) growth (Stock and Watson 1988).

The objective of this project is to empirically evaluate and extend the Stock and Watson (S&W) model. First, we will verify the assumptions of one-factor models, specifically by testing the tetrad

constraints in a dynamic settings. Then, we will asses its predictive performance by replicating the model on more recent data and comparing the estimated latent factor to actual GDP growth. Furthermore, because macroeconomic data frequently exhibits heavy tailed behaviors, we will test the robustness of the Kalman filter by simulating economies under progressively more elliptical distribution.

Finally, we will expand the number of indicators included in the model by incorporating more macroeconomic variables. To avoid adding too many variables, and therefore too much noise to the model, we will apply a Lasso penalty to the factor loadings of the indicators. Specifically, we will force sparsity on the new indicators, while simultaneously not assigning any penalty to the original four indicators in order to retain them. With this setup, we will ensure that the original four indicators will still represent the foundation of our model, while squeezing extra predictive value out of a larger dataset.

1. The Stock & Watson model

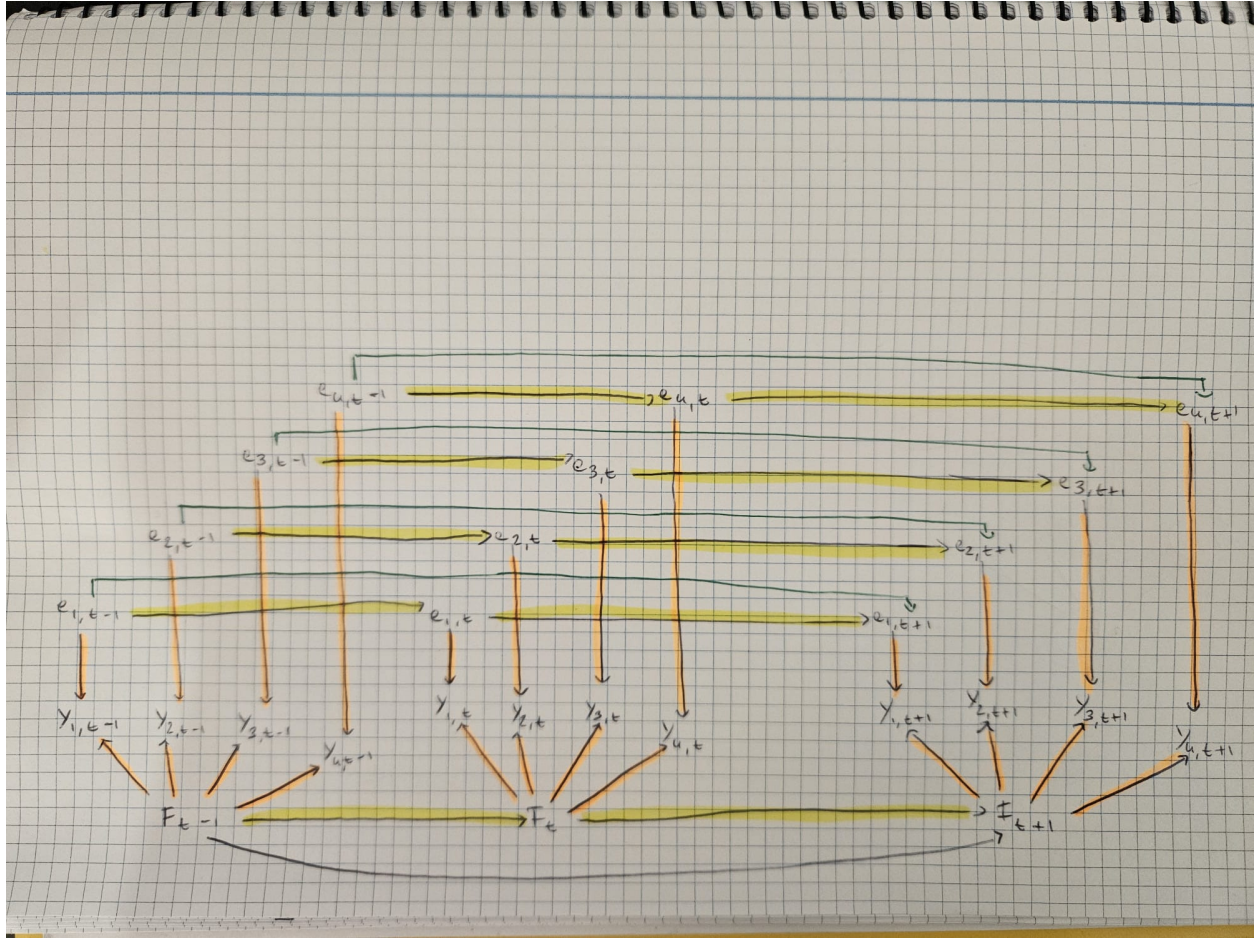
We first replicate model proposed by Stock and Watson. Their model assumes that the comovements of four macroeconomic indicators are driven by a single, unobserved latent factor which represents the overall business cycle. Formally, the model is represented by the following structural equations:

- The measurement equation: $y_t = \lambda f_t + e_t$
- The factor transition equation: $f_t = \phi_1 f_{t-1} + \phi_2 f_{t-2} + w_t$, $w_t \sim i.i.d.N(0, 1)$
- The error transition equation: $e_t = \psi_1 e_{t-1} + \psi_2 e_{t-2} + \epsilon_t$, $\epsilon_t \sim i.i.d.N(0, \sigma_i^2)$

Where:

- y_t is the (4×1) vector of normalized log-differences of the indicators at time t
- λ is the (4×1) vector of coefficients
- f_t is the (1×1) factor scalar at time t
- e_t is the (4×1) vector of errors at time t

Noticeably, to capture the cyclic patter of business cycles, both the factor and the error follow an $AR(2)$ process. Visually, the model can be represented by the following graphical model:



1.1. Latent factor estimation

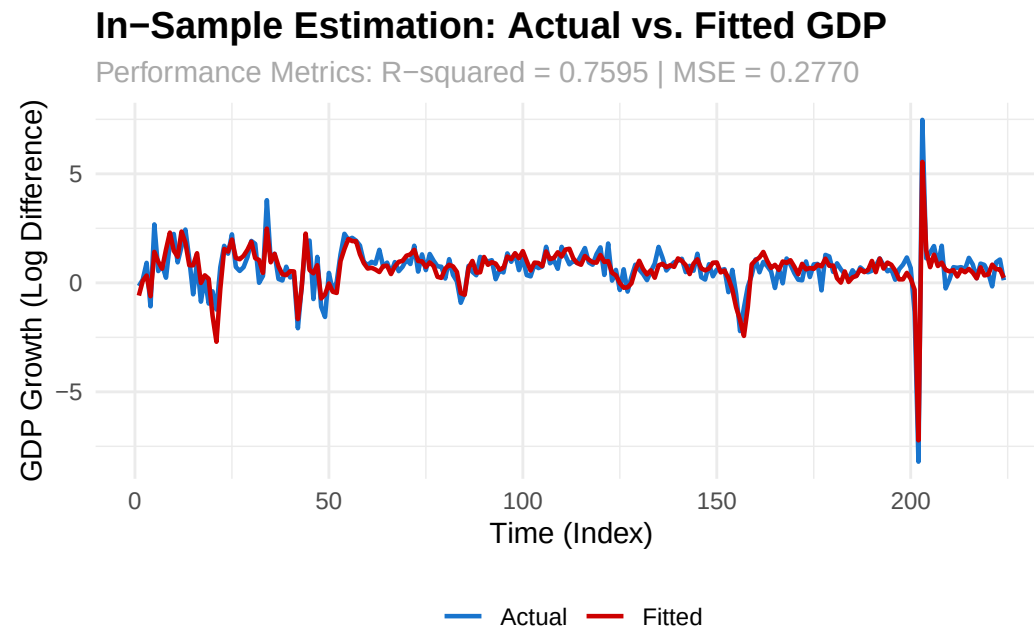
Given that the model is linear and that shocks w_t and w_t are assumed to be Gaussian and independent, the optimal estimator for the latent factor is given by the Kalman filter (Kalman 1960). However, this estimation technique requires the errors of the measurement equation e_t to be independent white noise, differently from the $AR(2)$ specification of the S&W model. To overcome this limitation, we take the $AR(2)$ errors out of the measurement equation and include them into the latent state vector, alongside the current and lagged values of the factor. The structural equations simplify to:

- The measurement equation: $y_t = Hh_t$
- The factor transition equation: $h_t = Fh_{t-1} + v_t, \quad v_t \sim i.i.d.N(0, \sigma)$

Where h_t is the state vector which now includes the current and lagged values of both the factor and the error. With this structure, the F matrix handles the $AR(2)$ dynamics of both the factor and the error.

1.2. Mapping the latent factor to GDP growth

After estimating the latent factor via the Kalman filter, we must rescale it to make it interpretable in terms of actual economic activity. To do so, we regress GDP onto the estimated factor by Ordinary Least Squares (OLS). This allows us to plot the estimated latent factor against GDP growth and calculate in-sample performance metrics, such as the coefficient of determination (R^2) and the Mean Squared Error (MSE).



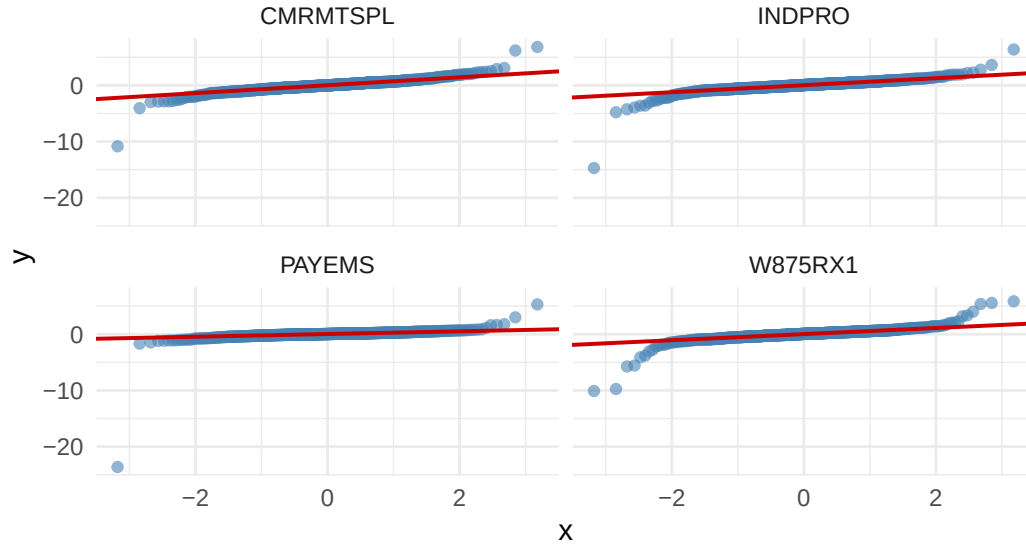
The empirical results demonstrate the strong explanatory power of the model proposed by Stock & Watson. The mapped factor explains approximately 76% of the total variance of the actual GDP growth ($R^2 = 0.7595$). Moreover, the model yields an in-sample MSE of 0.277, translating to a Root Mean Squared Error (RMSE) of roughly 0.526 percentage points. This RMSE might seem elevated, but it's important to note that our dataset includes periods of extreme macroeconomic volatility, such as the 2008 and COVID-19 crises, which inflate the error metrics.

1.3. The normality assumption

The dynamic factor model proposed by Stock and Watson explicitly assumes that all structural shocks are purely Gaussian. However, economic cycles frequently feature sudden and massive shocks (e.g. the COVID-19 crisis), hence macroeconomic indicators are rarely normally distributed. Instead, they tend to exhibit heavy-tailed behavior. As illustrated in the Quantile-Quantile (Q-Q) plot below, the distribution of our indicators indeed deviates from normality, confirming the presence of fat tails.

Normal Q–Q Plot for Macro Indicators

Points deviating from the red line indicate non-Gaussian 'fat tails'



Despite the violation of the normality assumption, the Kalman filter extracts a consistent estimate of the latent factor. Indeed, when the parameters are estimated via Gaussian quasi-maximum likelihood, consistency of the parameter estimates is preserved under non-Gaussianity, though with loss of efficiency in finite samples as the distribution moves away from Gaussianity and gets increasingly elliptical (White 1982). Chapter 2 will be dedicated to an empirical demonstration of the robustness of the Kalman filter, by simulating the economic model proposed by S&W under increasingly elliptical distributions.

1.4 Verifying that there is only one underlying factor (TO FINISH)

However, from our class we know that tetrad constraints to verify that there is only one underlying factor still hold under more elliptical distributions. Note, however, that tetrad constraints cannot be identified in a static way anymore, because the one factor model is dynamic. However, we can still verify them on the autocovariance structure (expand the explanation).

Table 1: Empirical Dynamic Tetrad Constraints

Lag	Tetrad 1	Tetrad 2	Tetrad 3
Lag 0	0.0524	0.0732	0.0208
Lag 1	0.0414	0.0770	0.0356
Lag 2	-0.0116	-0.0046	0.0070
Lag 3	-0.0054	-0.0031	0.0023
Lag 4	0.0066	0.0065	-0.0001
Lag 5	0.0005	0.0006	0.0001
Lag 6	0.0036	0.0041	0.0004

2. Breaking Gaussianity

The Kalman filter is derived under the assumption that all structural shocks follow a Gaussian distribution, which grants it the property of being the minimum variance unbiased estimator of the latent factor. When this assumption is violated, the filter estimated via Gaussian quasi-maximum likelihood remains consistent, meaning the estimated parameters still converge to their true values as $T \rightarrow \infty$, but **loses efficiency in finite samples**.

The reason for this is that under a misspecified likelihood, the Kalman gain is no longer optimal; therefore, the filter requires more observations to achieve the same precision as it would under correctly specified Gaussian shocks. This efficiency loss is particularly relevant for macroeconomic data, which with fewer than 700 observations can never be confidently characterized as truly Gaussian, and empirically exhibits heavy tailed behavior consistent with a student-t distribution (**geweke1993?**). We evaluate this by simulating the S&W data generating process under three distributional assumptions of increasing complexity: Gaussian, student-t, and skew-t, the latter of which introduces nonzero skewness via a slant parameter $\alpha \neq 0$.

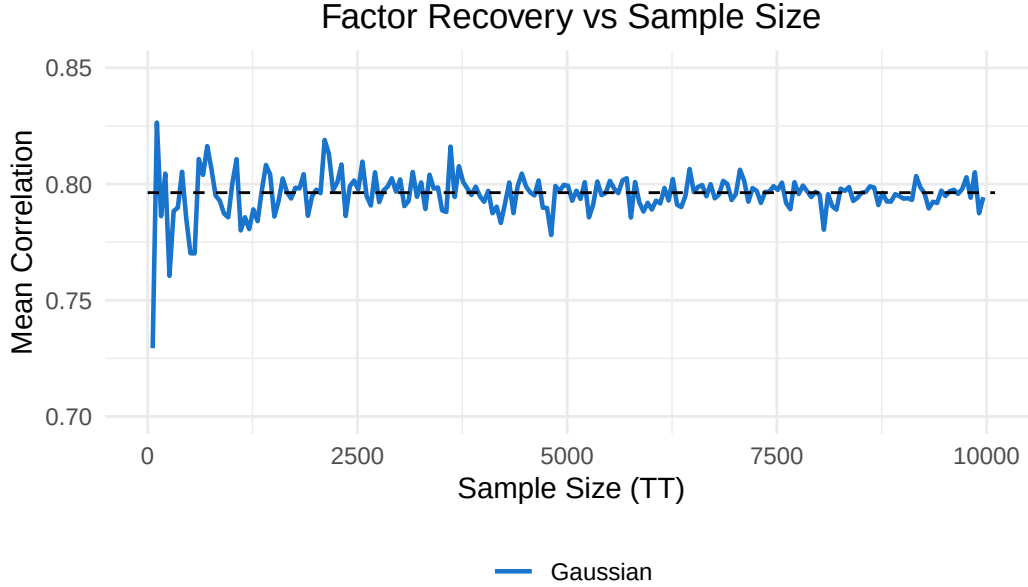
2.1 Gaussian Distribution

To simulate Gaussian data into our Kalman filter, we generate a structured array of various sample sizes to evaluate how the estimator performs in finite samples and what happens to its performance as $T \rightarrow \infty$.

For each sample size, we generate a true latent factor following an AR(2) process with coefficient values of $\hat{\phi}_1 = 0.293$ and $\hat{\phi}_2 = -0.098$, estimated from the S&W model in (**kalman-filter?**), with Gaussian shocks by assumption. We then construct the observable data by multiplying the true factor by the estimated factor loadings from (**kalman-filter?**) and adding normally distributed idiosyncratic errors, replicating the structure of the S&W measurement equation.

The simulated observables are normalized and fed into the Kalman filter, and the correlation between the estimated factor $\hat{f}_t$ and the true factor f_t is recorded across all sample sizes.

We first run a single iteration across sample sizes generated from `seq(10, 10000, by = 50)`, such that the lower bound captures realistic finite sample sizes representative of macroeconomic datasets, while the upper bound approaches the asymptotic nature of a dataset, where the property of consistency is expected to hold.



Dashed line indicates asymptotic factor recovery: 0.7963

Figure 1: Factor Recovery vs Sample Size under Gaussian Shocks

We see from Figure 1 that factor recovery under Gaussian shocks is stable across all sample sizes, with correlation between the estimated and true factor converging to approximately 0.7963 as T grows. The relatively high recovery even at small sample sizes is consistent with the Kalman filter being the minimum variance estimator under correctly specified Gaussian shocks, confirming that no efficiency loss is present when the distributional assumption holds.

2.2 Student-t Distribution

To break the assumption of Gaussianity, we simulate the S&W data generating process under a student-t distribution, which is well documented as a realistic characterization of finite sample macroeconomic data (geweke1993?).

Theoretically, given the QMLE consistency result, we should expect the Kalman filter to remain consistent in its recovery of the latent factor, meaning the estimates converge to the true factor as $T \rightarrow \infty$. However, we expect to observe an efficiency loss in finite samples, as the Kalman gain is derived under the Gaussian likelihood and is therefore suboptimal when the true distribution has heavier tails (White 1982; DurbinKoopman2012?).

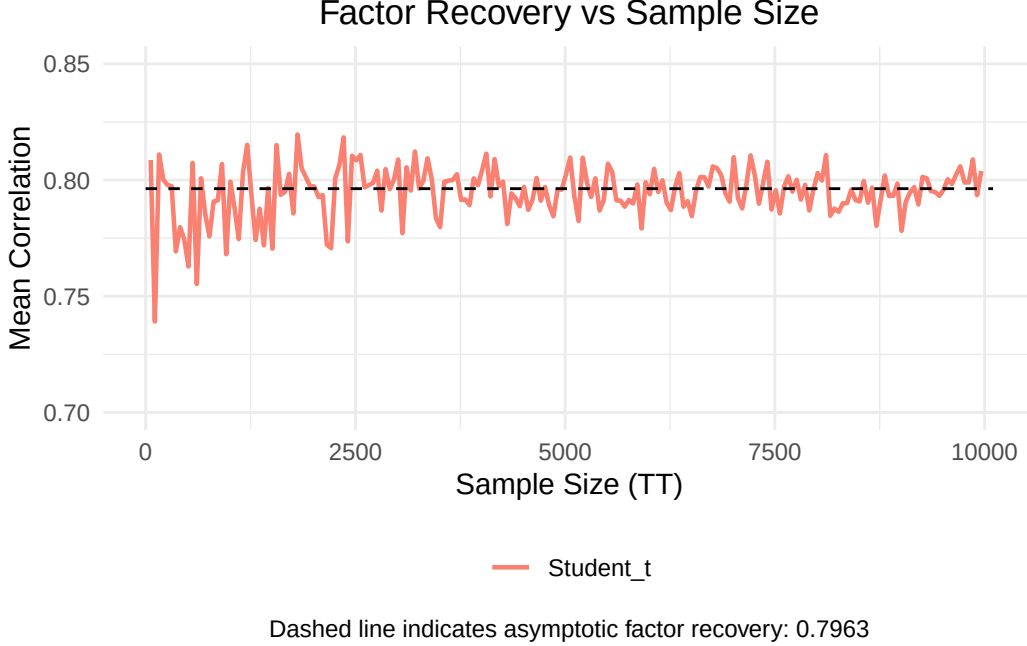


Figure 2: Factor Recovery vs Sample Size under Student-t Shocks

As illustrated in Figure 2, the Kalman filter remains consistent under student-t shocks, with factor recovery converging to approximately 0.7963 as $T \rightarrow \infty$, in line with the Gaussian baseline from (gaussian-dist?).

However, the trajectory is notably more volatile across sample sizes, particularly in the finite sample realm where the correlation fluctuates significantly before stabilizing. This is consistent with the theoretical prediction that the Kalman filter loses efficiency under heavy tailed distributions, as the misspecified Gaussian likelihood produces a suboptimal Kalman gain that requires more observations to average out the influence of extreme draws.

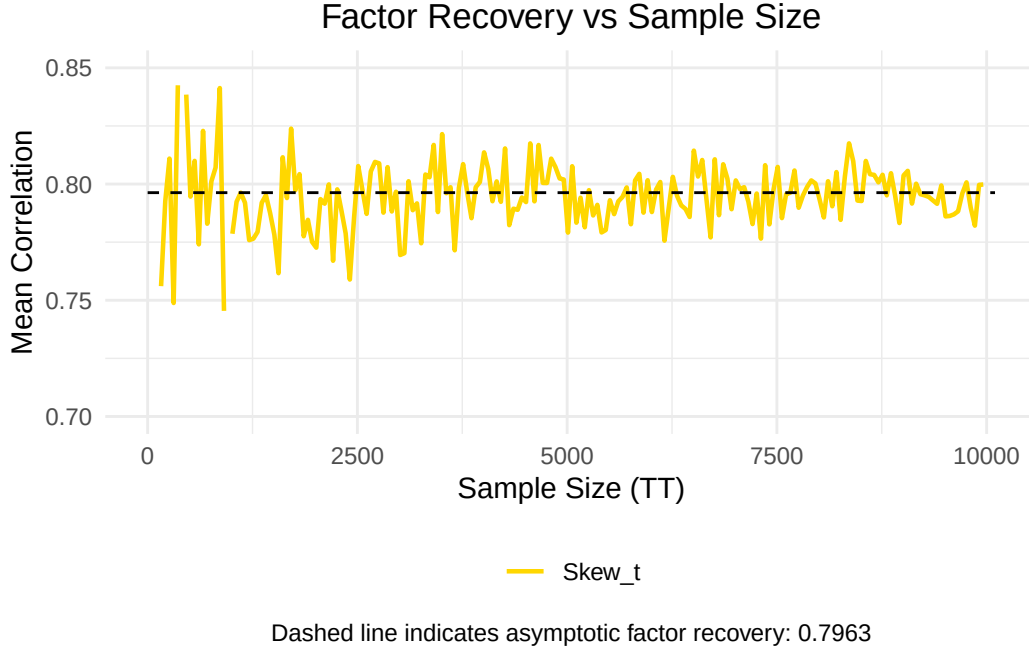
In other words, the fat tails of the student-t distribution introduce large idiosyncratic shocks that the filter struggles to separate from the common factor signal at small T , resulting in noisier factor recovery relative to the Gaussian case.

2.3 Skewed-Elliptical

For a more complex distributional assumption, we simulate the S&W data generating process under a skew-t distribution, which extends the student-t by introducing asymmetry via a nonzero slant parameter $\alpha \neq 0$.

Specifically, we set `alpha = 5` which induces a positive skew, meaning the distribution has a longer right tail relative to the symmetric student-t case. Note that our simulated factor shocks are generated as follows:

The factor shocks are recentered by subtracting the sample mean, as the skew-t distribution has a nonzero mean when $\alpha \neq 0$, which would otherwise violate the zero mean assumption of the Kalman filter and introduce a systematic bias into the factor estimates.



Dashed line indicates asymptotic factor recovery: 0.7963

Figure 3: Factor Recovery vs Sample Size under Skew-t Shocks

Indeed, from our results in Figure 3 we see that the Kalman filter remains consistent under skew-t shocks, with factor recovery converging to approximately 0.7963 as $T \rightarrow \infty$, in line with both the Gaussian and student-t baselines. This confirms the QMLE consistency result, as the estimator recovers the true latent factor regardless of the distributional assumption placed on the shocks.

However, the trajectory is notably more volatile relative to both the Gaussian and student-t cases, with correlation fluctuating between 0.75 and 0.85 even at large sample sizes. This persistent volatility reflects the additional efficiency cost introduced by the nonzero skewness of the shock distribution, as the positive slant parameter $\alpha = 5$ generates asymmetric extreme draws that the Kalman filter, operating under a misspecified Gaussian likelihood, struggles to average out even at large T .

2.4 Monte Carlo Simulation

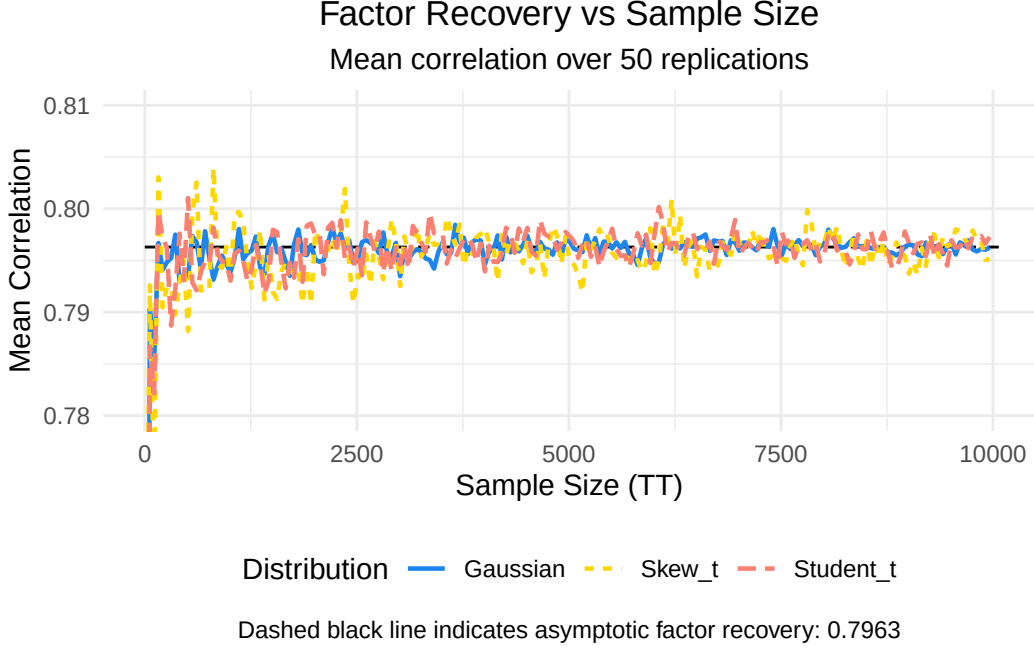


Figure 4: Factor Recovery vs Sample Size, Averaged Across 50 MC Simulation

As illustrated in Figure 4, the three distributions converge to the same asymptotic factor recovery correlation of approximately 0.7963 as $T \rightarrow \infty$, directly confirming the QMLE consistency result. At small sample sizes, all three distributions exhibit greater volatility and slightly lower recovery, with the skew-t and student-t lines deviating more from the Gaussian baseline, consistent with the theoretical prediction of finite sample efficiency loss under non-Gaussianity.

Notably, both the student-t and skew-t distributions exhibit greater volatility relative to the Gaussian baseline, particularly at small sample sizes, with the skew-t remaining the most volatile throughout. It should be noted that the y-axis range is intentionally narrow to highlight these differences, and as reported in Table 2, the absolute magnitude of the fluctuations is small in practice, with standard deviations of 0.021 and 0.027 for the student-t and skew-t, respectively. Nevertheless, all three distributions converge to the same asymptotic recovery value of 0.7963, confirming the QMLE consistency result regardless of the distributional assumption.

2.5 Efficiency Loss under Non-Gaussian Data

Table 2: Factor Recovery: Mean and Variance across 200 Replications

Distribution	Mean	Variance	SD
Gaussian	0.796293	0.000208	0.014438
Student-t	0.792082	0.000430	0.020724
Skew-t	0.794169	0.000718	0.026802

To formally demonstrate the efficiency loss under non-Gaussian distributions, we run 200 replications of the Monte Carlo simulation at the empirical sample size of $T = 675$, corresponding to the number of observations in our FRED dataset, and compute the variance of factor recovery across replications.

As reported in Table 2, the Gaussian estimator achieves the lowest variance of 0.000208, confirming its optimality under correctly specified shocks. The student-t estimator exhibits a variance of 0.000430, roughly twice that of the Gaussian case, while the skew-t estimator yields the highest variance of 0.000718, approximately 3.5 times the Gaussian baseline. Importantly, the mean recovery is nearly identical across all three distributions at approximately 0.794, confirming that the estimator remains consistent regardless of the distributional assumption.

Together, these results provide formal evidence of the consistency without efficiency property of the Gaussian quasi-maximum likelihood Kalman filter: the estimator converges to the correct value under all three distributions, but requires significantly more observations to do so with the same precision when shocks are non-Gaussian.

3. Two-Factor Model

motivation: want to improve performance (increase R^2 from 75%)

what happens when we generate a second latent variable by including more macro variables in the model? (I.e., we add in financial variables and generate financial latent factor) and we try to run S&W under the one-factor assumption? Should be biased, breaks; we try to recover the second latent factor with some sort of dynamic parallel analysis.

tld: - find vars to generate second variable - build new model using vars, run thru kalman filter - compare to results from part 1.1, show that it is biased - explain why this is - try to extract second latent factor using parallel analysis, need to show this in section 1 before Kalman filters

3.1 Introducing More Variables

3.2 Estimated Using Stock & Watson

3.3 Parallel Analysis

Bibliography

- Kalman, Rudolph Emil. 1960. "A New Approach to Linear Filtering and Prediction Problems." *Transactions of the ASME-Journal of Basic Engineering* 82 (Series D): 35–45.
- Stock, James H, and Mark W Watson. 1988. "A Probability Model of the Coincident Economic Indicators." Working Paper 2772. National Bureau of Economic Research. <https://www.nber.org/papers/w2772>.
- White, Halbert. 1982. "Maximum Likelihood Estimation of Misspecified Models." *Econometrica: Journal of the Econometric Society* 50 (1): 1–25.